

Summer Bootcamp – Session 3

Corrections to the Optimization Exercises

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Correction Guidelines

The solutions follow the numbering of the exercise sheet. Necessary conditions are not used as sufficient conditions unless convexity, curvature, or a direct comparison supplies the missing argument. Numerical experiments use the seeds stated below and stop when $\|\nabla f(x_k)\|_2 \leq 10^{-8}$, or, for a constrained problem, when both feasibility and stationarity residuals are at most 10^{-8} .

The companion file `session_3_optimization_corrections.py` contains the complete reference computations for Exercises 10 and 11. The separate Portfolio Optimization Project is not corrected in this document; its status is stated at the end.

1 One-Dimensional Unconstrained Optimization

Exercise 1 – Critical points are only candidates. For $f_1(x) = x^4 - 4x^2 + 3$,

$$f'_1(x) = 4x(x^2 - 2), \quad f''_1(x) = 12x^2 - 8.$$

Thus 0 is a strict local maximum and $\pm\sqrt{2}$ are strict local minima. Since $f_1(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, the two minima are global:

$$f_1(0) = 3, \quad f_1(\pm\sqrt{2}) = -1.$$

For $f_2(x) = x^3 - 3x + 1$, the interior critical points are ± 1 , with $f'_2(x) = 6x$. The boundary comparison is essential:

x	-2	-1	1	2
$f_2(x)$	-1	3	-1	3

Hence the global minima on $[-2, 2]$ are $x = -2$ and $x = 1$, while the global maxima are $x = -1$ and $x = 2$. Among the interior points, -1 is a strict local maximum and 1 a strict local minimum. Relative to the closed domain, the two endpoints are also one-sided local extrema.

Finally, for $f_3(x) = \log(1 + e^x) - \frac{3}{4}x$,

$$f'_3(x) = \frac{e^x}{1 + e^x} - \frac{3}{4}, \quad f''_3(x) = \frac{e^x}{(1 + e^x)^2} > 0.$$

The function is strictly convex, and stationarity gives $e^x = 3$. Moreover, $f_3(x) \rightarrow +\infty$ at both ends of the real line. Therefore

Result

$$x_3^* = \log 3, \quad f_3(x_3^*) = \log 4 - \frac{3}{4} \log 3,$$

and this minimizer is unique and global.

The sketches must show the symmetric double well of f_1 , the cubic on its restricted interval for f_2 , and the strictly convex bowl of f_3 , with the points just identified marked explicitly.

Exercise 2 – A parameter changes the number of local optima. Stationarity for $f_a(x) = \frac{1}{4}x^4 - \frac{a}{2}x^2 + x$ is

$$p_a(x) := f'_a(x) = x^3 - ax + 1 = 0.$$

The turning points of this cubic occur at $x = \pm s$, where $s = \sqrt{a/3}$, and

$$p_a(-s) = 1 + \frac{2a^{3/2}}{3\sqrt{3}} > 0, \quad p_a(s) = 1 - \frac{2a^{3/2}}{3\sqrt{3}}.$$

Introduce the threshold

$$a_c = \left(\frac{3\sqrt{3}}{2} \right)^{2/3} = \frac{3}{2^{2/3}}.$$

Then p_a has one real root for $0 < a < a_c$, three distinct real roots for $a > a_c$, and a simple root plus a double root when $a = a_c$.

In the three-root regime, write the roots $r_1 < r_2 < r_3$. The sign pattern is

x	$(-\infty, r_1)$	r_1	(r_1, r_2)	r_2	(r_2, r_3)	r_3	(r_3, ∞)
$f'_a(x)$	–	0	+	0	–	0	+

so r_1, r_3 are strict local minima and r_2 is a strict local maximum. At $a = a_c$, the negative simple root is a strict local minimum; the positive double root of p_a is stationary but is neither a minimum nor a maximum.

Because $f_a(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, a global minimum exists. Checking $f'_a = 0$ only generates its candidates. A rigorous comparison can evaluate f_a at all stationary points; at a stationary root r , the identity $r^3 = ar - 1$ simplifies the value to

$$f_a(r) = -\frac{a}{4}r^2 + \frac{3}{4}r.$$

There is also a direct argument in the three-root regime: for every $x > 0$, $f_a(-x) = f_a(x) - 2x < f_a(x)$. Hence the positive local minimum cannot be global, and the negative local minimum r_1 is the unique global minimizer.

Exercise 3 – Newton’s method as a local model. Here

$$f'(x) = e^x - 3 + \frac{x}{2}, \quad f''(x) = e^x + \frac{1}{2} > 0.$$

Strict convexity and coercivity give one global minimizer. It is the unique root of f' , and it may be written using the Lambert W function as

$$x^* = 6 - W(2e^6).$$

Newton’s iteration is

Result

$$x_{k+1} = x_k - \frac{e^{x_k} - 3 + x_k/2}{e^{x_k} + 1/2}.$$

Indeed, the quadratic model in a displacement p is

$$m_k(p) = f(x_k) + f'(x_k)p + \frac{1}{2}f''(x_k)p^2,$$

whose unique minimizer is $p_k = -f'(x_k)/f''(x_k)$. At every nonstationary x_k ,

$$f'(x_k)p_k = -\frac{f'(x_k)^2}{f''(x_k)} < 0,$$

so p_k is a descent direction. This does not guarantee that the full step $\alpha_k = 1$ decreases the true nonlinear objective. Armijo backtracking replaces it by $x_{k+1} = x_k + \alpha_k p_k$, repeatedly reducing α_k until

$$f(x_k + \alpha_k p_k) \leq f(x_k) + c\alpha_k f'(x_k)p_k.$$

It therefore preserves the fast local Newton step when it is acceptable and forces sufficient decrease when the quadratic model is poor.

2 Multidimensional Unconstrained Optimization

Exercise 4 – Quadratic optimization and conditioning. For $f(x) = \frac{1}{2}x^\top Qx - c^\top x$, with $Q = Q^\top \succ 0$,

$$\nabla f(x) = Qx - c, \quad \nabla^2 f(x) = Q.$$

For distinct x, y and $t \in (0, 1)$, direct expansion gives

$$tf(x) + (1-t)f(y) - f(tx + (1-t)y) = \frac{1}{2}t(1-t)(x-y)^\top Q(x-y) > 0,$$

so f is strictly convex. Positive definiteness also implies $\ker Q = \{0\}$, hence Q is invertible. The stationarity system has the unique solution

$$x^\star = Q^{-1}c,$$

where a numerical implementation should use a linear solve. Completing the square yields the global certificate

$$f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top Q(x - x^\star) \geq 0.$$

Let $Q = U \operatorname{diag}(\lambda_1, \dots, \lambda_d) U^\top$. In coordinates $z = U^\top(x - x^\star)$, the objective gap is $\frac{1}{2} \sum_i \lambda_i z_i^2$: the eigenvectors are principal directions and the eigenvalues are their curvatures.

For gradient descent, $e_k = x_k - x^\star$ satisfies

$$e_{k+1} = (I - \alpha Q)e_k, \quad (U^\top e_{k+1})_i = (1 - \alpha \lambda_i)(U^\top e_k)_i.$$

Convergence for every initial point is equivalent to

$$0 < \alpha < \frac{2}{\lambda_{\max}}.$$

When $\kappa(Q) \approx 1$, level sets are nearly spherical and all coordinates contract at comparable rates. When $\kappa(Q) \gg 1$, the level sets are elongated: a step small enough for the largest curvature makes slow progress along the smallest-curvature direction. The constant step $2/(\lambda_{\min} + \lambda_{\max})$ has worst-coordinate contraction factor $(\kappa - 1)/(\kappa + 1)$.

Exercise 5 – Least squares, ridge regression, and conditioning. For $\ell(x) = \frac{1}{2} \|Ax - b\|_2^2$,

$$\nabla \ell(x) = A^\top(Ax - b),$$

so every minimizer solves the normal equations

$$A^\top Ax = A^\top b.$$

The solution is unique exactly when $\ker A = \{0\}$, equivalently when A has full column rank. At any solution $x^\star$,

$$A^\top(b - Ax^\star) = 0,$$

so the residual is orthogonal to the column space of A ; $Ax^\star$ is the orthogonal projection of b onto that space.

For ridge regression,

$$\nabla \ell_\lambda(x) = (A^\top A + \lambda I)x - A^\top b,$$

and therefore

Result

$$x_\lambda = (A^\top A + \lambda I)^{-1} A^\top b.$$

For every nonzero v ,

$$v^\top (A^\top A + \lambda I)v = \|Av\|_2^2 + \lambda \|v\|_2^2 > 0.$$

Thus ridge has a unique solution even if A is rank deficient.

If σ_i are the singular values of A , the Hessian eigenvalues become $\sigma_i^2 + \lambda$. Hence

$$\kappa(A^\top A + \lambda I) = \frac{\sigma_{\max}^2 + \lambda}{\sigma_{\min}^2 + \lambda},$$

where $\sigma_{\min} = 0$ in a rank-deficient problem. Regularization raises small curvatures and improves stability, but the singular-vector coefficients are shrunk by factors $\sigma_i/(\sigma_i^2 + \lambda)$. This creates bias in exchange for lower sensitivity to noise.

Exercise 6 – Second-order conditions in several dimensions. For f_1 ,

$$\nabla f_1 = \begin{pmatrix} 2x + y - 4 \\ x + 2y + 2 \end{pmatrix}, \quad H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The stationary point is $(10/3, -8/3)$, and the Hessian eigenvalues 1, 3 are positive. It is the unique global minimizer. For f_2 , stationarity requires

$$4x(x^2 - 1) = 0, \quad 4y(y^2 - 2) = 0,$$

so $x \in \{0, \pm 1\}$ and $y \in \{0, \pm\sqrt{2}\}$. Moreover,

$$H_2(x, y) = \text{diag}(12x^2 - 4, 12y^2 - 8).$$

The classification is

points	Hessian signs	classification
$(\pm 1, \pm\sqrt{2})$	$(+, +)$	strict global minima, value -5
$(0, 0)$	$(-, -)$	strict local maximum, value 0
$(\pm 1, 0)$	$(+, -)$	saddles
$(0, \pm\sqrt{2})$	$(-, +)$	saddles

There are no degenerate stationary candidates for the stated f_2 : both diagonal Hessian entries are nonzero at every candidate. Thus the requested directional inspection is not needed here; this observation is itself the appropriate check.

For f_3 , the only stationary point is $(0, 0)$, and

$$H_3 = \begin{pmatrix} 2 & 2 \\ 2 & -2 \end{pmatrix}, \quad \det H_3 = -8 < 0.$$

It is a saddle. The restrictions $f_3(x, 0) = x^2$ and $f_3(0, y) = -y^2$ give the same conclusion and show that the function is unbounded above and below. Contour plots should respectively display an elliptic bowl, four equivalent wells for f_2 , and hyperbolic saddle contours for f_3 .

3 Multidimensional Constrained Optimization

Exercise 7 – Projection onto an affine constraint. The projection is the solution of

$$\min_x \frac{1}{2} \|x - z\|_2^2 \quad \text{subject to} \quad a^\top x = b.$$

With $\mathcal{L}(x, \lambda) = \frac{1}{2} \|x - z\|_2^2 + \lambda(a^\top x - b)$, stationarity gives $x = z - \lambda a$. Enforcing feasibility yields

Result

$$\text{proj}_C(z) = z - \frac{a^\top z - b}{\|a\|_2^2} a.$$

The correction vanishes when $z \in C$, and $z - \text{proj}_C(z) \in \text{span}(a)$, the normal space of the hyperplane.

For $C = \{x : Ax = b\}$, use the vector multiplier λ . Stationarity and feasibility give

$$x = z - A^\top \lambda, \quad AA^\top \lambda = Az - b.$$

If A has full row rank, then $A^\top y = 0$ implies $y = 0$, and $y^\top AA^\top y = \|A^\top y\|^2 > 0$ for $y \neq 0$. Thus $AA^\top$ is invertible and

$$\boxed{\text{proj}_C(z) = z - A^\top (AA^\top)^{-1} (Az - b).}$$

Exercise 8 – Equality constraints, duality, and a saddle point. Set $S = AQ^{-1}A^\top$. The Lagrangian and KKT system are

$$\begin{aligned} \mathcal{L}(x, \lambda) &= \frac{1}{2} x^\top Q x + c^\top x + \lambda^\top (Ax - b), \\ \begin{pmatrix} Q & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x^* \\ \lambda^* \end{pmatrix} &= \begin{pmatrix} -c \\ b \end{pmatrix}. \end{aligned}$$

Since $Q \succ 0$ and A has full row rank, $S \succ 0$. Eliminating x gives

Result

$$\begin{aligned} \lambda^* &= -S^{-1}(b + AQ^{-1}c), \\ x^* &= -Q^{-1}c + Q^{-1}A^\top S^{-1}(b + AQ^{-1}c). \end{aligned}$$

For a fixed multiplier, minimizing the Lagrangian in x at $x(\lambda) = -Q^{-1}(c + A^\top \lambda)$ gives

$$q(\lambda) = -\frac{1}{2}(c + A^\top \lambda)^\top Q^{-1}(c + A^\top \lambda) - b^\top \lambda.$$

Its Hessian is $-S \prec 0$, so the dual function is strictly concave.

If x is feasible, then for every λ ,

$$q(\lambda) = \inf_y \mathcal{L}(y, \lambda) \leq \mathcal{L}(x, \lambda) = f(x),$$

which is weak duality. At the KKT pair, x^* minimizes $\mathcal{L}(\cdot, \lambda^*)$ and is feasible, hence

$$q(\lambda^*) = \mathcal{L}(x^*, \lambda^*) = f(x^*).$$

This proves strong duality and the saddle relations

$$\mathcal{L}(x^*, \lambda) = \mathcal{L}(x^*, \lambda^*) \leq \mathcal{L}(x, \lambda^*) \quad \text{for all } x, \lambda.$$

Finally, if $V(b)$ is the optimal value, the envelope theorem with the stated sign convention gives

$$DV(b)[\Delta b] = -\lambda^{*\top} \Delta b, \quad \nabla_b V(b) = -\lambda^*.$$

Exercise 9 – Projection onto a half-space with KKT. For the inequality $g(x) = a^\top x - b \leq 0$,

$$\mathcal{L}(x, \lambda) = \frac{1}{2} \|x - z\|^2 + \lambda(a^\top x - b).$$

The KKT conditions are

$$x - z + \lambda a = 0, \quad a^\top x \leq b, \quad \lambda \geq 0, \quad \lambda(a^\top x - b) = 0.$$

If $a^\top z \leq b$, the unconstrained minimizer is feasible and $(x^*, \lambda^*) = (z, 0)$. If $a^\top z > b$, the constraint is active and

$$\lambda^* = \frac{a^\top z - b}{\|a\|^2}, \quad x^* = z - \frac{a^\top z - b}{\|a\|^2} a.$$

Consequently,

Result

$$\text{proj}_{\{a^\top x \leq b\}}(z) = z - \frac{(a^\top z - b)_+}{\|a\|_2^2} a.$$

The positive part switches continuously from an inactive to an active constraint at the boundary. The objective is strictly convex, the constraint is affine, and a strictly feasible point exists because $a \neq 0$; therefore the KKT conditions are sufficient (and the optimizer is unique).

4 Python Algorithms

Exercise 10 – Implement gradient descent from scratch. A reusable implementation must evaluate the stopping test before taking the next step and must keep the initial point in its history. The essential loop is shown below; the companion Python file adds input checks and all requested plots.

```
def gradient_descent(f, grad, x0, step_rule, tol=1e-8, max_iter=25_000):
    x = np.asarray(x0, dtype=float).copy()
    xs, values, norms, steps = [], [], [], []
    for _ in range(max_iter + 1):
        value, g = float(f(x)), np.asarray(grad(x), dtype=float)
        xs.append(x.copy()); values.append(value)
        norms.append(np.linalg.norm(g))
        if norms[-1] <= tol:
            break
        alpha = float(step_rule(f, x, g))
        steps.append(alpha)
        x = x - alpha * g
    return {"x": x, "iterates": np.asarray(xs),
            "values": np.asarray(values),
            "gradient_norms": np.asarray(norms),
            "step_sizes": np.asarray(steps)}

def armijo_step(f, x, g, initial=1.0, c=1e-4, rho=0.5):
    alpha, fx = initial, f(x)
    while f(x - alpha*g) > fx - c*alpha*(g @ g):
        alpha *= rho
    return alpha
```

For the quadratic experiment, seed 3 gives

$$\lambda_{\min} = 0.115303501, \quad \lambda_{\max} = 74.677637749, \quad \kappa(Q) = 647.661.$$

The benchmark is computed with `np.linalg.solve(Q, c)`, never by explicitly forming Q^{-1} .

Step	Iterations	Final $\ \nabla f\ $	$\ x - x^*\ $	Status
$1/\lambda_{\max}$	11199	9.99×10^{-9}	8.66×10^{-8}	converged
$2/(\lambda_{\min} + \lambda_{\max})$	5656	9.99×10^{-9}	7.37×10^{-8}	converged
$2.05/\lambda_{\max}$	7358	overflow	—	unstable

The second stable step roughly halves the iteration count. The third violates the stability bound and eventually amplifies the largest-eigenvalue component.

For Rosenbrock,

$$\nabla r(x, y) = \begin{pmatrix} -400x(y - x^2) - 2(1 - x) \\ 200(y - x^2) \end{pmatrix}.$$

At $(-1.2, 1)$, a centered finite-difference check with $h = 10^{-6}$ has infinity-norm error 2.24×10^{-8} . Armijo backtracking with $c = 10^{-4}$, $\rho = 1/2$, and initial trial step 1 reaches

$$x = (0.99999999, 0.99999998), \quad r(x) = 6.11 \times 10^{-17}, \quad \|\nabla r(x)\| = 1.00 \times 10^{-8}$$

after 19435 accepted steps. Slow motion along the narrow curved valley is expected for plain steepest descent.

The reference script produces the four diagnostics requested in the exercise: the quadratic objective gap, the quadratic gradient norm, the path in the extreme eigendirections of Q , and the Rosenbrock path over contours. A correct plot uses a positive floor before taking a logarithm of an objective gap at machine precision.

Exercise 11 – Implement primal–dual gradient descent–ascent. The exact reference pair is obtained from

$$\begin{pmatrix} Q & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x^* \\ \lambda^* \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}.$$

For $e_k = x_k - x^*$ and $d_k = \lambda_k - \lambda^*$, the simultaneous iteration has transition matrix

$$\begin{pmatrix} e_{k+1} \\ d_{k+1} \end{pmatrix} = \underbrace{\begin{pmatrix} I - \alpha Q & -\alpha A^\top \\ \beta A & I \end{pmatrix}}_{T_{\alpha, \beta}} \begin{pmatrix} e_k \\ d_k \end{pmatrix}.$$

Thus the linear iteration converges for every initial condition exactly when $\rho(T_{\alpha, \beta}) < 1$. This spectral-radius calculation explains the observed behavior more precisely than a visual judgment from the trajectories.

Using seed 11, $d = 20$, $m = 3$, and the construction in the exercise:

(α, β)	$\rho(T)$	behavior	iterations	max(final residuals)
(0.02, 0.02)	0.988187	stable	1621	9.92×10^{-9}
(0.03, 0.03)	1.064405	growing oscillations	3038	diverged

The Lagrangian value need not decrease monotonically: one variable descends while the multiplier ascends. Feasibility and stationarity residuals, not the Lagrangian alone, certify convergence.

For the augmented Lagrangian,

$$\nabla_x \mathcal{L}_\rho = Qx + c + A^\top \lambda + \rho A^\top (Ax - b),$$

so the primal block in the transition matrix becomes $I - \alpha(Q + \rho A^\top A)$. With $(\alpha, \beta) = (0.005, 0.02)$, the reproducible comparison is

ρ	spectral radius	iterations	feasibility	stationarity
0.1	0.996814	5770	6.84×10^{-10}	9.98×10^{-9}
1	0.996803	5766	7.70×10^{-10}	9.99×10^{-9}
10	0.998026	9916	2.40×10^{-10}	1.00×10^{-8}

Larger ρ penalizes infeasibility more strongly but also increases primal curvature, so a step size that is effective at $\rho = 1$ need not remain effective at $\rho = 10$. All converged solutions agree with the exact KKT solution to the reported tolerance.

For the optional extragradient method, define

$$F(x, \lambda) = \begin{pmatrix} Qx + c + A^\top \lambda \\ -(Ax - b) \end{pmatrix}.$$

Then compute

$$\tilde{z}_k = z_k - \eta F(z_k), \quad z_{k+1} = z_k - \eta F(\tilde{z}_k).$$

The look-ahead evaluation damps the rotational behavior created by the off-diagonal $A, A^\top$ blocks and often permits a wider stable step range than simultaneous descent–ascent. Stability should still be verified from residual histories or the corresponding linear iteration matrix.

5 Portfolio Optimization

Exercise 12 – Risk, return, and the Sharpe ratio. For deterministic weights w , linearity of expectation and covariance give

$$\mathbb{E}[w^\top R] = w^\top \mu, \quad \text{Var}(w^\top R) = \mathbb{E}[(w^\top (R - \mu))^2] = w^\top \Sigma w.$$

For GMV, use $\mathcal{L}(w, \lambda) = \frac{1}{2} w^\top \Sigma w + \lambda(\mathbf{1}^\top w - 1)$. Stationarity gives $w = -\lambda \Sigma^{-1} \mathbf{1}$, and the budget fixes the multiplier. Thus

Result

$$w_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}.$$

Let $e = \mu - r_f \mathbf{1}$. Cauchy-Schwarz in the Σ -inner product gives

$$e^\top w = (\Sigma^{-1/2}e)^\top (\Sigma^{1/2}w) \leq \sqrt{e^\top \Sigma^{-1}e} \sqrt{w^\top \Sigma w}.$$

Equality holds precisely in the tangency direction $w \propto \Sigma^{-1}e$. If $D = \mathbf{1}^\top \Sigma^{-1}e \neq 0$, full investment fixes

$$w_T = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{D}.$$

For the usual maximum-positive-Sharpe interpretation one also checks the sign of D ; a negative normalization reverses the direction.

Equal weight is a transparent, low-estimation benchmark. GMV uses covariance but not expected returns and is appropriate when risk control is primary. Tangency targets risk-adjusted excess return but is especially sensitive to mean estimates. With long-only constraints the closed forms can contain negative weights and become infeasible. For long-only GMV, for example, KKT requires

$$\Sigma w + \lambda \mathbf{1} - \nu = 0, \quad \mathbf{1}^\top w = 1, \quad w \geq 0, \quad \nu \geq 0, \quad \nu_i w_i = 0.$$

An analogous stationarity equation, together with the same primal, dual, and complementarity conditions, is required for the chosen tangency formulation.

Exercise 13 – Derive the efficient frontier. Let

$$C = \begin{pmatrix} \mathbf{1}^\top \\ \mu^\top \end{pmatrix}, \quad d_m = \begin{pmatrix} 1 \\ m \end{pmatrix}.$$

The Lagrangian gives the KKT system

$$\begin{pmatrix} \Sigma & C^\top \\ C & 0 \end{pmatrix} \begin{pmatrix} w \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ d_m \end{pmatrix}.$$

Eliminating w , and assuming the two constraints are independent, yields

Result

$$w(m) = \Sigma^{-1} C^\top (C \Sigma^{-1} C^\top)^{-1} d_m.$$

Define

$$A_0 = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}, \quad B_0 = \mathbf{1}^\top \Sigma^{-1} \mu, \quad D_0 = \mu^\top \Sigma^{-1} \mu, \quad \Delta = A_0 D_0 - B_0^2 > 0.$$

Since the multiplier calculation gives $w(m)^\top \Sigma w(m) = d_m^\top (C \Sigma^{-1} C^\top)^{-1} d_m$, the frontier variance is

Result

$$\sigma^2(m) = \frac{A_0 m^2 - 2B_0 m + D_0}{\Delta}.$$

It is a convex quadratic in target return. In mean–standard-deviation space it produces a hyperbola; the efficient branch begins at the GMV return $m_{\text{GMV}} = B_0/A_0$ and moves toward higher returns.

Long-only constraints and caps make the feasible set polyhedral. As m changes, weights hit or leave bounds; each fixed active set has its own quadratic formula, producing piecewise behavior and visible active-set changes. Before tracing a frontier, code should verify symmetry and positive semidefiniteness of Σ , preferably positive definiteness for uniqueness; rank and consistency of the equality constraints; feasibility of the budget and target; bounds and caps; solver success; and final KKT residuals. Under long-only full investment, a necessary and sufficient return check without caps is $m \in [\min_i \mu_i, \max_i \mu_i]$.

Exercise 14 – Estimation error and robust weights. Write $q = \hat{\Sigma}^{-1}(\hat{\mu} - r_f \mathbf{1})$. Keeping covariance and the risk-free rate fixed gives the exact perturbation formula

$$q(\hat{\mu} + \varepsilon h, \hat{\Sigma}) = q(\hat{\mu}, \hat{\Sigma}) + \varepsilon \hat{\Sigma}^{-1} h.$$

Therefore

$$D_\mu q[h] = \hat{\Sigma}^{-1} h, \quad J_\mu q = \hat{\Sigma}^{-1}.$$

If $\hat{\Sigma} = U \text{diag}(\lambda_1, \dots, \lambda_d) U^\top$ and $h = \sum_i h_i u_i$, then

$$\Delta q = \varepsilon \sum_i \frac{h_i}{\lambda_i} u_i, \quad \|\Delta q\|_2 \leq \frac{|\varepsilon|}{\lambda_{\min}(\hat{\Sigma})} \|h\|_2.$$

Small-eigenvalue directions amplify mean error the most. The corresponding relative bound is the right-hand side divided by $\|q\|_2$, when $q \neq 0$.

For normalized weights $w = q/(\mathbf{1}^\top q)$, set $D = \mathbf{1}^\top q$. Differentiation gives

$$D_\mu w[h] = \frac{D \hat{\Sigma}^{-1} h - q \mathbf{1}^\top \hat{\Sigma}^{-1} h}{D^2}.$$

The denominator explains the additional instability when D is close to zero.

GMV weights do not contain $\hat{\mu}$, so they are immune to mean estimation error. They still contain both $\hat{\Sigma}^{-1}$ and a normalizing denominator, and hence remain sensitive to covariance error, especially near singularity.

For ridge regularization, the eigenvectors are unchanged and

$$\lambda_i(\hat{\Sigma}_\delta) = \lambda_i + \delta, \quad \kappa(\hat{\Sigma}_\delta) = \frac{\lambda_{\max} + \delta}{\lambda_{\min} + \delta}.$$

This condition number decreases toward 1 as δ increases, although the estimator becomes more biased toward an isotropic covariance.

In-sample predicted volatility reuses the observations that selected both the covariance and the weights, so it is optimistically biased. Out-of-sample realized volatility must be computed from untouched future observations. Expected returns are particularly noisy, and optimization applies an inverse covariance to that noise; the resulting extreme weights can make return-based objectives much less stable than GMV.

A transparent selection rule is:

1. split the training period chronologically into an inner estimation sample and a validation sample;
2. declare a grid such as $\delta = \tau \text{tr}(\hat{\Sigma})/d$ before inspecting the final test set;
3. fit each candidate on the inner sample and rank it by validation realized variance, optionally with a predeclared concentration or turnover tie-breaker;
4. choose δ , refit using all training observations, and evaluate the final test sample exactly once.

Rolling validation may replace a single split, provided every validation fold occurs strictly after its estimation window and the final test set remains untouched.

Separate Portfolio Optimization Project

The correction to the separate Portfolio Optimization Project is still being drafted. It is intentionally not included in this correction document.